A Survey of Causal Representation Learning and Causal Discovery in Deep Learning

InteractiveSurvey

Abstract

The rapid advancement of deep learning has revolutionized pattern recognition, yet standard neural architectures often function as black boxes that fail to generalize under distributional shifts or support counterfactual reasoning due to their inability to capture underlying causal mechanisms. Consequently, integrating causal inference principles with representation learning has become imperative for developing robust, interpretable artificial intelligence systems capable of distinguishing invariant structures from environment-specific noise. This survey provides a comprehensive examination of the intersection between causal representation learning and causal discovery, focusing on methodologies that explicitly model structural dependencies and causal directions among latent factors within deep learning frameworks. We synthesize recent theoretical breakthroughs and algorithmic innovations to elucidate how modern techniques leverage interventions, distribution shifts, and structural constraints to achieve identifiability in nonlinear settings where classical approaches fail. Key findings highlight the critical role of multi-environment variability and temporal nonstationarity as inductive biases for latent recovery, while demonstrating how sparsity regularization and causal abstraction enable disentangled representations that separate content from style. Furthermore, the review evaluates the application of these frameworks to counterfactual generative modeling and algorithmic fairness, addressing challenges related to latent confounders and intervention fidelity. By providing a unified taxonomy of current approaches and critically assessing the trade-offs between model flexibility and identifiability guarantees, this work serves as a foundational resource for researchers aiming to develop next-generation AI systems capable of robust generalization and fair decision-making.

1 Introduction

The rapid advancement of deep learning has revolutionized pattern recognition and generative modeling, yet standard neural architectures often function as black boxes that capture spurious correlations rather than underlying causal mechanisms. While these models excel at minimizing prediction error on independent and identically distributed data, they frequently fail to generalize under distributional shifts or support counterfactual reasoning, which are essential for robust artificial intelligence. The inability to distinguish between invariant causal structures and environment-specific noise limits the deployment of deep learning systems in safety-critical domains such as healthcare and autonomous driving. Consequently, there is a growing imperative to integrate causal inference principles with representation learning, aiming to recover latent variables that correspond to true generative factors of the data [1].

This survey paper provides a comprehensive examination of the intersection between causal representation learning and causal discovery within deep learning frameworks [2]. Unlike traditional approaches that treat latent variables as statistically independent components, this work focuses on methodologies that explicitly model the structural dependencies and causal directions among latent factors. We explore how modern techniques leverage interventions, distribution shifts, and structural constraints to achieve identifiability in nonlinear settings where classical independent component analysis fails. By synthesizing recent theoretical breakthroughs and algorithmic innovations, this review elucidates the pathways through which deep generative models can learn disentangled, interpretable, and manipulable representations grounded in causal semantics.

The first major theme of this review addresses the critical role of environmental variability and in-

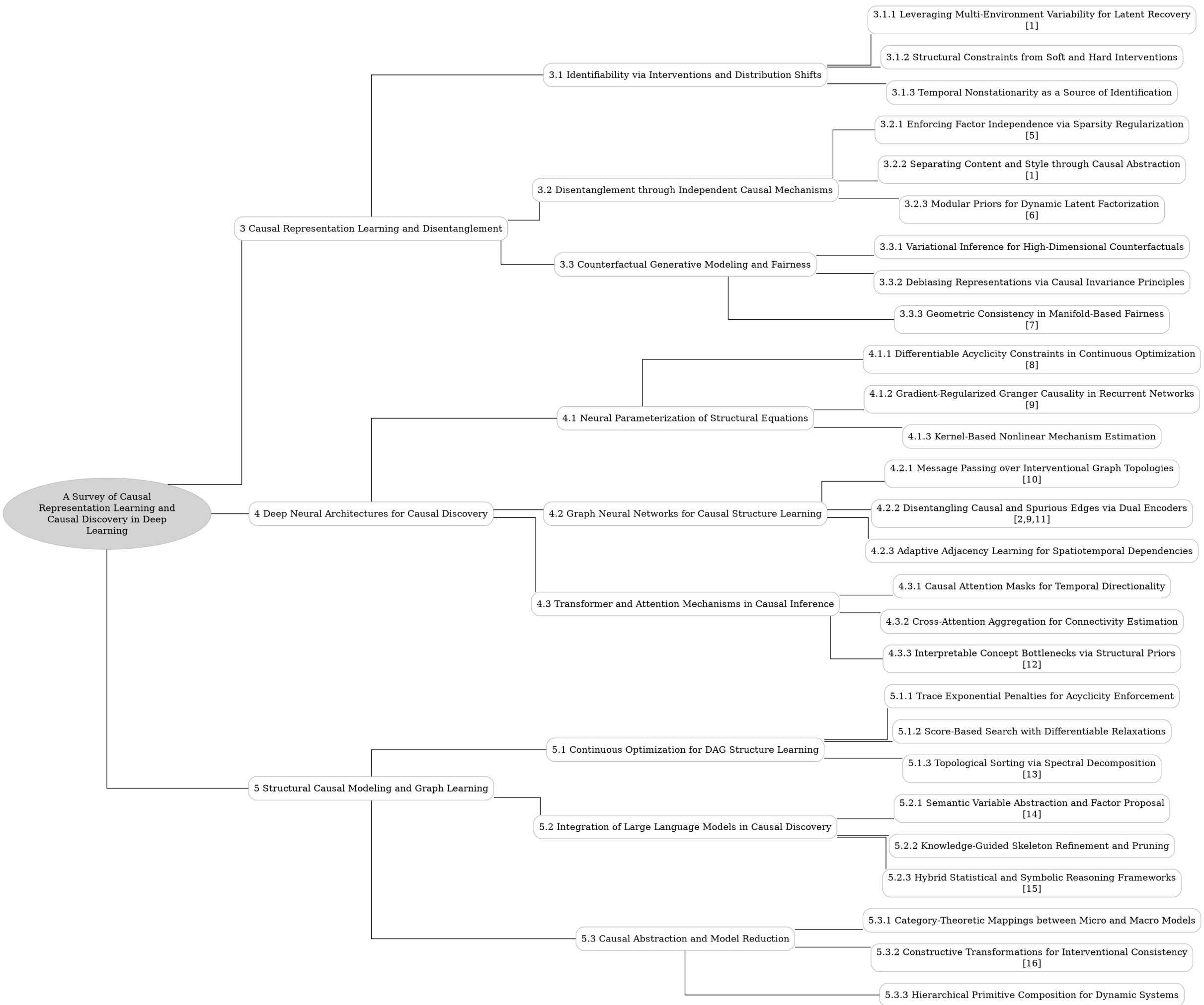


Figure 1: The outline of our survey: A Survey of Causal Representation Learning and Causal Discovery in Deep Learning

$$P(\boldsymbol{x}_i) = \prod_{j=1}^{D} P(\boldsymbol{x}_{ij} \mid \mathrm{pa}_{\mathsf{B}}(j)).$$

Figure 2: Chart from 'Towards Interpretable Deep Generative Models via Causal Representation Learning' (arXiv:2504.11609)

$$X_i^c = A_i^T X_i$$

Figure 3: Chart from 'Group Interventions on Deep Networks for Causal Discovery in Subsystems' (arXiv:2510.23906)

tervention types in identifying latent causal structures. We analyze how multi-environment data serves as a powerful inductive bias, treating domain shifts as soft interventions that expose invariant mechanisms while allowing marginal distributions to vary. The discussion extends to the theoretical distinctions between hard and soft interventions, evaluating how each imposes different structural constraints on the learning process. Furthermore, we examine the utilization of temporal nonstationarity as an auxiliary signal, demonstrating how time-varying distributions can resolve ambiguities in latent recovery without requiring explicit knowledge of intervention targets, thereby transforming dynamic instability into a constructive source of identification.

Subsequently, we delve into strategies for achieving disentanglement through the principle of Independent Causal Mechanisms and structured priors. This section details how sparsity regularization breaks rotational symmetries in factor analysis, enforcing modular representations where specific latent sources influence distinct subsets of observations. We further explore causal abstraction techniques that separate content from style by modeling directional asymmetries, enabling block-identifiability even when mutual independence does not hold. The review also covers modular priors for dynamic latent factorization, which encode causal dependencies directly into the prior distribution of variational autoencoders. These methods facilitate the capture of intricate generative processes where factors influence one another, supporting counterfactual reasoning and enhancing interpretability beyond simple compression.

Finally, the survey investigates the application of these causal frameworks to counterfactual generative modeling and algorithmic fairness [3]. We discuss how variational inference is adapted to approximate intractable posteriors in high-dimensional spaces, enabling the estimation of counterfactuals from purely observational data. The analysis highlights challenges related to latent confounders and the fidelity of generated outcomes, reviewing recent solutions that incorporate orthogonality constraints and mixture oracles to ensure robust interventional predictions. By connecting theoretical identifiability guarantees with practical generative capabilities, this section underscores the potential of causal representation learning to mitigate bias and support reliable decision-making in complex, real-world scenarios.

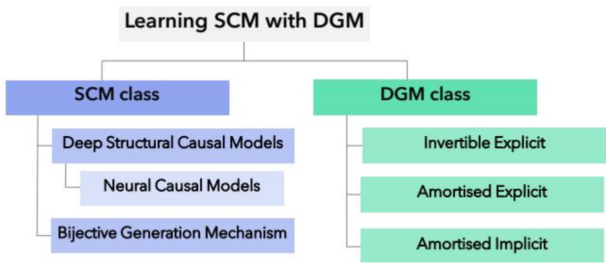


Figure 4: Chart from 'Learning Structural Causal Models through Deep Generative Models: Methods, Guarantees, and Challenges' (arXiv:2405.05025)

The primary contribution of this survey lies in its systematic synthesis of diverse methodologies that bridge the gap between unsupervised representation learning and rigorous causal inference [1]. We provide a unified taxonomy of current approaches, categorizing them by their reliance on intervention types, structural assumptions, and temporal dynamics. Beyond summarizing existing literature, this work critically evaluates the trade-offs between model flexibility and identifiability guarantees, offering insights into the limitations of current techniques in high-dimensional settings. By clarifying the theoretical foundations and practical implications of causal deep learning, this paper serves as a foundational resource for researchers aiming to develop next-generation AI systems capable of robust generalization, interpretable reasoning, and fair decision-making [4].

$$m : \mathcal{D} \rightarrow \mathcal{Y},$$

Figure 5: Chart from 'Navigating causal deep learning: a living paper' (arXiv:2212.00911)

2 Causal Representation Learning and Disentanglement

2.1 Identifiability via Interventions and Distribution Shifts

2.1.1 Leveraging Multi-Environment Variability for Latent Recovery

Leveraging multi-environment variability serves as a critical inductive bias for identifying latent causal structures where independent and identically distributed data fails. By treating distinct domains, experimental conditions, or temporal contexts as soft interventions on underlying mechanisms, these approaches exploit distributional shifts to disentangle confounded factors. The core premise relies on the assumption that while the marginal distributions of latent variables change across environments, the fundamental causal relationships and structural equations remain invariant, thereby providing the necessary constraints to recover unique latent representations up to trivial ambiguities.

Methodologically, this paradigm often extends variational autoencoder frameworks to incorporate environment-specific parameters or auxiliary variables that modulate the latent prior. Techniques such as minimizing score differences across distributions or enforcing independence between latent codes and environment indices allow models to isolate exogenous noise terms from endogenous causal variables. Recent advances have moved beyond strict intervention requirements, utilizing rank deficiency constraints and nonparametric assumptions to identify latent graphs even when the number of variables or intervention targets is unknown, effectively transforming the problem of causal discovery into one of representation learning under heterogeneous conditions [1].

Despite theoretical guarantees of identifiability under sufficient environmental diversity, practical implementation faces significant optimization challenges, particularly in high-dimensional settings with complex causal topologies. Models often struggle with latent space collapse or fail to converge when the overlap assumption in the latent support is violated, necessitating specialized regularization terms like negative entropy or maximum mean discrepancy to maintain manifold coverage. Consequently, current research focuses on developing robust architectures that can scale to realistic datasets while preserving the causal interpretability of the learned latent factors without relying on explicit knowledge of the intervention targets.

2.1.2 Structural Constraints from Soft and Hard Interventions

Hard interventions, which sever all incoming causal edges to a target variable, provide strong structural constraints that facilitate the identification of latent causal graphs. To achieve statistical identifiability without imposing restrictive functional constraints, recent theoretical frameworks often mandate multiple distinct hard interventions per latent node. Specifically, ensuring "interventional discrepancy," where interventional distributions differ almost everywhere, allows for the recovery of the underlying structural causal model up to permutation. However, this approach relies on the strong assumption that variables can be completely decoupled from their parents, a condition frequently violated in real-world systems where variables may be immutable or safety-critical.

In contrast, soft interventions modify the functional mechanism of a variable while preserving its dependency on a subset of original parents, offering a more flexible but theoretically challenging alternative. Because soft interventions do not remove direct causal links, the identifiable structure is generally limited to the transitive closure or ancestral relationships rather than the precise directed acyclic graph. Consequently, recovering the exact causal topology under soft interventions typically requires additional parametric assumptions, such as linearity in the latent structural equations or specific forms of nonlinear mixing functions, to compensate for the weaker structural signals compared to perfect interventions.

The trade-off between intervention type and identifiability dictates the necessary constraints on the learning framework. While hard interventions enable nonparametric identification under relatively mild conditions, soft interventions often necessitate strict assumptions regarding the faithfulness of interventional changes or the sparsity of mechanism shifts. Recent advances attempt to bridge this gap by leveraging paired interventional data or specific structural restrictions on the latent graph, such as pure-child constraints. Ultimately, the choice between soft and hard intervention models determines the extent to which the causal structure can be disentangled from observational ambiguities, balancing the realism of the intervention model against the rigor of the resulting identifiability guarantees.

2.1.3 Temporal Nonstationarity as a Source of Identification

Temporal nonstationarity serves as a critical auxiliary signal for achieving identifiability in latent causal models, effectively overcoming the inherent unidentifiability of standard nonlinear independent component analysis. By leveraging distribution shifts across time, methods can distinguish between invariant causal mechanisms and changing noise distributions or intervention targets. This approach posits that while the underlying structural equations remain constant, the statistical properties of exogenous noise or the specific variables subjected to intervention vary temporally, providing the necessary constraints to recover the true latent representation up to trivial indeterminacies.

Recent theoretical advancements demonstrate that observing a system under unknown, atomic interventions within a temporal sequence allows for the identification of causal variables without requiring explicit knowledge of the intervention targets. Frameworks operating in this regime utilize the conditional independence structures induced by these temporal shifts to disentangle causally related factors that would otherwise remain entangled under static observational conditions. Specifically, by modeling the latent dynamics as a nonstationary process, it becomes possible to establish component-wise identifiability through the analysis of Jacobian transformations and the varying support of the latent distributions across different time segments.

Furthermore, this paradigm extends to heterogeneous environments where the sparse mechanism shift hypothesis holds, enabling causal discovery even when the exact nature of the nonstationarity is unobserved. Unlike approaches relying on independent priors or fully supervised labels, techniques exploiting temporal nonstationarity provide provable guarantees for nonlinear latent hierarchical models under weak supervision. These methods effectively transform the challenge of time-varying distributions from a confounding factor into a constructive source of information, facilitating the recovery of robust causal representations capable of supporting counterfactual reasoning and generalization across diverse contexts.

2.2 Disentanglement through Independent Causal Mechanisms

2.2.1 Enforcing Factor Independence via Sparsity Regularization

Enforcing factor independence through sparsity regularization constitutes a pivotal strategy for achieving identifiability in latent variable models, particularly when statistical independence alone proves insufficient. By imposing sparsity-inducing priors on the loading matrix or structural parameters, these methods constrain the solution space to favor interpretable, modular representations where each latent factor influences only a distinct subset of observed variables. This structural sparsity effectively breaks the rotational symmetry inherent in standard factor analysis, ensuring that the recovered factors correspond to unique generative mechanisms rather than arbitrary linear combinations.

The theoretical underpinning of this approach relies on the assumption that true causal mechanisms are sparse, meaning specific latent sources exclusively generate particular subsets of data features. Regularization terms, often formulated as ℓ_1 penalties on the Jacobian of the decoding function or the adjacency matrix of the latent graph, explicitly penalize dense connectivity. This encourages the learning algorithm to discover a factorization where the conditional dependencies between latent variables and observations are minimized, aligning the learned representation with the Independent Causal Mechanisms principle [5]. Consequently, the model avoids entangled solutions by forcing the network to isolate distinct factors of variation through sparse connectivity patterns.

Recent advancements extend these concepts to nonlinear settings, leveraging mechanism sparsity to guarantee identifiability up to permutation and element-wise scaling. Unlike traditional methods that assume fully independent latent components, sparsity-regularized approaches accommodate structured dependencies among factors while maintaining disentanglement at the mechanism

$$H_N^p = \frac{1}{|\mathcal{W}_p|} \sum_{k \in \mathcal{W}_p} \mathbb{I}[R_k^p \leq N].$$

Figure 6: Chart from 'INTERPRETABLE CAUSAL REPRESENTATION LEARNING FOR BIOLOGICAL DATA IN THE PATHWAY SPACE' (arXiv:2506.12439)

level. By integrating these constraints into variational autoencoder frameworks or differentiable causal discovery pipelines, researchers can recover robust latent structures even in the presence of complex, non-Gaussian distributions, thereby bridging the gap between unsupervised representation learning and causal inference.

2.2.2 Separating Content and Style through Causal Abstraction

Separating content and style via causal abstraction relies on modeling the generative process as a Structural Causal Model where content variables causally influence style factors, but not vice versa. This directional asymmetry, often formalized as $c \to s$, allows for the definition of block-identifiability, distinguishing it from traditional independent component analysis which assumes mutual independence among all latent factors. By relaxing the independence constraint, these methods acknowledge that stylistic attributes often depend on the underlying semantic content, thereby capturing more realistic data manifolds while preserving the modularity required for meaningful intervention.

Theoretical frameworks in this domain leverage weak supervision or known causal graphs to achieve identification up to component-wise reparameterization. Unlike fully unsupervised approaches that suffer from non-identifiability without restrictive assumptions, causal abstraction utilizes auxiliary information, such as domain labels or paired observations, to disentangle the joint distribution. Techniques often employ variational inference with structured priors that enforce the specific causal topology between content and style blocks, ensuring that interventions on content propagate naturally to style while maintaining the integrity of the semantic representation.

Practical implementations typically utilize deep generative models equipped with mechanisms to simulate counterfactuals, verifying that style components remain invariant under perfect interventions on content when appropriate conditions are met. This capability facilitates robust transfer learning and domain adaptation, as the isolated content representation can be combined with novel style factors from different distributions. Consequently, causal abstraction provides a principled pathway to learn representations that are not only statistically distinct but also semantically manipulable, enabling precise control over generated outputs in complex, high-dimensional settings [1].

2.2.3 Modular Priors for Dynamic Latent Factorization

Modular priors for dynamic latent factorization address the limitations of traditional independent factor assumptions by explicitly encoding causal dependencies within the latent space. Unlike standard variational autoencoders that enforce statistical independence, these methods utilize Structural Causal Models (SCMs) to define a prior distribution where latent variables are nodes in a directed acyclic graph [6]. This approach factorizes the joint prior into a product of conditional distributions, $p(\mathbf{z}) = \prod p(z_i | \mathbf{z}_{\text{pa}_i}, u_i)$, allowing the model to capture intricate generative mechanisms where factors influence one another rather than varying in isolation.

$$\mathbf{P}^* = \arg\max_{\mathbf{P}} \sum_k \Big[\mathcal{L}(\phi, \theta = 1, k, \mathbf{P}) \\ + \lambda_2 \left\|\nabla_{\theta=1} \mathcal{L}(\phi, \theta = 1, k, \mathbf{P})\right\|^2 \Big].$$

Figure 7: Chart from 'Disentangled Representation Learning' (arXiv:2211.11695)

In dynamic settings, this modular structure is extended to incorporate temporal evolution and interventionist capabilities. The prior is often conditioned on auxiliary variables or previous time steps, enabling the disentanglement of causally related factors through non-stationary distributions or specific action sequences. By modeling the noise terms as independent exogenous variables while maintaining structural dependencies among the latents, these frameworks facilitate identifiability even in nonlinear regimes. This formulation supports counterfactual reasoning, allowing for precise interventions on specific latent dimensions while preserving the integrity of the remaining causal graph.

The implementation of such priors typically involves modifying the evidence lower bound to penalize deviations from the prescribed causal factorization rather than simple total correlation. Techniques range from parametric assumptions using exponential families to nonparametric approaches leveraging normalizing flows to map independent noise to dependent latent structures. These methods significantly enhance the interpretability of learned representations, moving beyond mere compression to uncovering the underlying modular machinery of complex, multi-modal data. Consequently, they provide a robust foundation for tasks requiring robust generalization under distributional shifts and explicit causal manipulation.

2.3 Counterfactual Generative Modeling and Fairness

2.3.1 Variational Inference for High-Dimensional Counterfactuals

Variational inference has emerged as a pivotal framework for approximating intractable posteriors in high-dimensional causal representation learning, particularly when estimating counterfactuals from observational data. By leveraging the variational autoencoder architecture, researchers embed structural causal models directly into the latent prior, enabling the simultaneous learning of disentangled causal factors and their underlying mechanisms. This approach replaces the traditional abduction step with an amortized inference network that maps high-dimensional observations to distributions over exogenous noise terms, effectively bypassing the computational bottlenecks associated with exact Bayesian inversion in complex, non-linear systems.

However, scaling these methods to high-dimensional spaces introduces significant challenges regarding identifiability and the fidelity of the learned latent structure. While explicit latent causal models successfully unify generative modeling with causal semantics, they often struggle to detect latent confounders that do not directly influence low-level data, leading to potential biases in counterfactual generation. Recent advancements address these limitations by incorporating orthogonality constraints and mixture oracles to ensure that the inferred latent variables correspond to true causal entities up to permutation and element-wise reparameterization, thereby enhancing the robustness of interventional predictions.

Furthermore, the integration of normalizing flows and diffusion processes within variational frameworks has refined the precision of counterfactual estimation by modeling complex, non-Gaussian noise distributions. These hybrid architectures facilitate a more accurate reconstruction of the abduction-action-prediction pipeline, allowing for reliable simulation of hypothetical scenarios even in the absence of controlled randomized trials. Despite these strides, the requirement that all relevant causal variables must be reflected in the observed data remains a fundamental constraint, necessitating future work on detecting unmeasured confounders through auxiliary graph discovery and invariant principle regularization.

2.3.2 Debiasing Representations via Causal Invariance Principles

Debasing representations through causal invariance principles addresses the critical limitation of standard deep learning models that rely on spurious correlations rather than underlying causal mechanisms. By enforcing the Principle of Independent Causal Mechanisms, these methods posit that the conditional distribution of an effect given its causes remains invariant across different environments or domains. This invariance serves as a robust signature for distinguishing true causal factors from confounding variables, ensuring that learned representations capture stable generative processes rather than dataset-specific artifacts.

The implementation of these principles typically involves multi-environment learning frameworks where the model is optimized to minimize prediction risk while maximizing the stability of latent conditional distributions across shifting data domains. Techniques often reparameterize latent variables as structural causal models, explicitly modeling independent noise terms to facilitate disentanglement. By treating the latent space as a set of causally related variables linked by a directed acyclic graph, algorithms can identify representations that satisfy conditional independence statements, thereby achieving component-wise identifiability even in the presence of complex, interacting latent factors.

Consequently, representations learned via causal invariance exhibit superior out-of-distribution generalization capabilities compared to traditional disentangled approaches that assume independent priors without causal structure. When underlying factors are causally dependent, standard independence assumptions fail, whereas invariance-based methods correctly recover the structural relationships by leveraging changes in the joint distribution induced by environmental shifts. This approach not only enhances model robustness against distributional shifts but also provides a theoretically grounded pathway toward interpretable machine learning systems that align with the true data-generating process.

2.3.3 Geometric Consistency in Manifold-Based Fairness

Geometric consistency in manifold-based fairness leverages the intrinsic topology of data distributions to ensure that learned representations preserve causal structures essential for equitable decision-making [7]. By modeling observed data as lying on a low-dimensional Riemannian manifold, these approaches posit that fair representations must respect the geodesic distances and local neighborhood relationships defined by the underlying causal mechanisms. This geometric perspective moves beyond standard statistical parity, requiring that the mapping from latent causal variables to observed data remains diffeomorphic, thereby maintaining the structural integrity of the data generation process across different demographic groups.

$$\mathcal{L}_{geo} = \mathbb{E}\Big[||G(z) - G(z_{cf})||_F + \beta \sum_{k=1}^{d_x} ||H_k(z) - H_k(z_{cf})||_F\Big]$$

Figure 8: Chart from 'Causal Manifold Fairness: Enforcing Geometric Invariance in Representation Learning' (arXiv:2601.03032)

The identifiability of such manifold-embedded causal variables relies heavily on assumptions regarding the smoothness and invertibility of the decoding function, often formalized through conditions on the Jacobian rank and conditional independence. Recent theoretical advancements demonstrate that under specific interventional faithfulness and support constraints, the latent causal graph can be recovered up to permutation and scaling, even in the presence of non-linear mixing. Crucially, enforcing geometric consistency acts as a regularizer that reduces the equivalence class of admissible solutions, effectively pinning down the true causal structure by eliminating spurious correlations that violate the manifold's intrinsic geometry.

Furthermore, integrating these geometric constraints into generative frameworks allows for the construction of counterfactuals that are both semantically meaningful and causally valid. When the learned representation aligns with the true data manifold, algorithmic recourse and fairness interventions can be performed by traversing geodesics that correspond to valid causal transitions rather than arbitrary perturbations. This ensures that fairness criteria based on causal quantities are robust to distributional shifts, as the geometric properties of the manifold provide a stable foundation for defining invariant features that are decoupled from sensitive attributes while retaining predictive utility [7].

3 Deep Neural Architectures for Causal Discovery

3.1 Neural Parameterization of Structural Equations

3.1.1 Differentiable Acyclicity Constraints in Continuous Optimization

Traditional causal structure learning relies on discrete combinatorial optimization, which becomes computationally intractable as the number of variables increases due to the super-exponential search space of directed acyclic graphs (DAGs) [8]. To overcome this, recent paradigms reformulate structure learning as a continuous optimization problem by relaxing the discrete acyclicity constraint into a smooth, differentiable equality constraint. This transformation allows the application of efficient gradient-based optimization techniques, such as augmented Lagrangian methods, to learn the adjacency matrix directly alongside structural parameters, effectively bridging the gap between discrete graph theory and continuous numerical optimization.

The cornerstone of this approach is the characterization of acyclicity via the trace of the matrix exponential. Specifically, for a weighted adjacency matrix W, the function $h(W) = \text{Tr}(e^{W \circ W}) - d$ serves as a necessary and sufficient condition for acyclicity, vanishing if and only if the graph contains no cycles. By incorporating this term as a penalty in the loss function, the optimization process can navigate the continuous parameter space while strictly enforcing the DAG property without requiring explicit cycle-checking al-

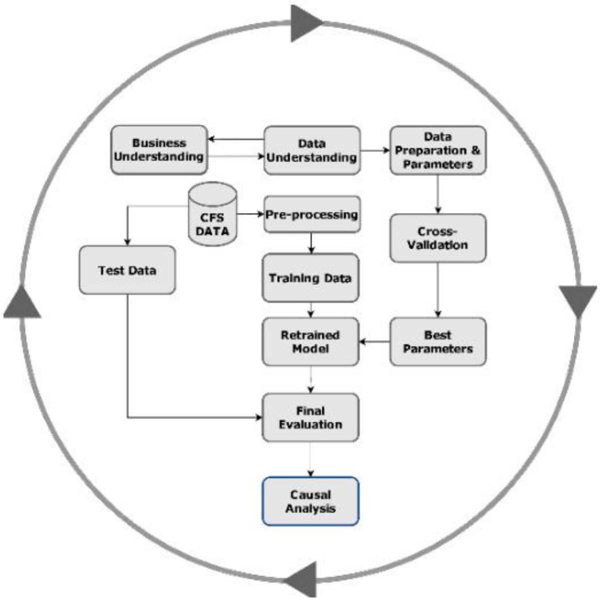


Figure 9: Chart from 'Causal Analysis of Customer Churn Using Deep Learning' (arXiv:2304.10604)

gorithms at every iteration. This differentiable formulation ensures that the resulting graph structure remains valid throughout the training trajectory, facilitating end-to-end learning within deep neural network architectures.

Despite its theoretical elegance, this continuous relaxation introduces challenges regarding numerical stability and the landscape of the optimization objective. The non-convex nature of the trace exponential function can lead to local minima or slow convergence, particularly in high-dimensional settings with complex nonlinear dependencies. Consequently, subsequent refinements have explored alternative characterizations, such as spectral constraints or polynomial approximations, to improve computational efficiency and robustness. These advancements collectively establish differentiable acyclicity constraints as a fundamental mechanism for scalable causal discovery, enabling the integration of structural priors into modern machine learning pipelines.

3.1.2 Gradient-Regularized Granger Causality in Recurrent Networks

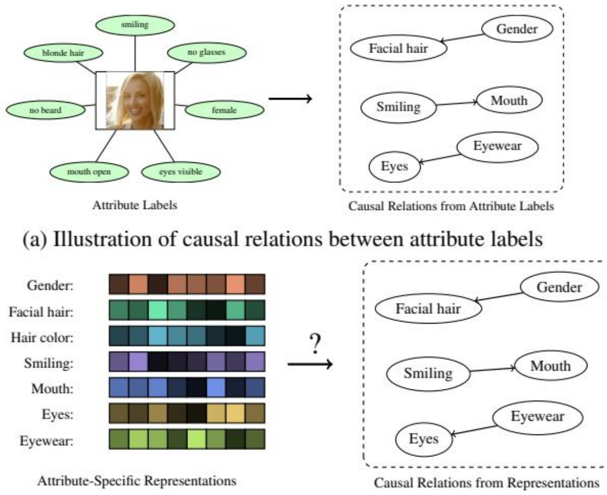
Gradient-Regularized Granger Causality in Recurrent Networks addresses the challenge of inferring nonlinear causal dependencies in high-dimensional time series by integrating sparsity-inducing penalties directly into the optimization of recurrent architectures. Unlike traditional linear Granger causality, which relies on vector autoregressive models, this approach leverages the representational capacity of Gated Recurrent Units or Long Short-Term Memory networks to capture complex temporal dynamics. The core mechanism involves augmenting the standard prediction loss with a regularization term derived from the gradients of the network inputs, effectively forcing the model to learn sparse connectivity patterns that reflect true causal drivers while suppressing spurious correlations arising from shared latent confounders.

The technical implementation typically employs group lasso or similar norm-based constraints on the input-to-hidden weight matrices, ensuring that only variables with significant predictive influence maintain non-zero gradient pathways. By penalizing the magnitude of input gradients during backpropagation, the method encourages the recurrent unit to ignore irrelevant historical information, thereby mimicking the conditional independence tests fundamental to causal discovery. This gradient-based regularization is particularly effective in scenarios where the number of variables exceeds the sample size, as it prevents overfitting and promotes the recovery of a parsimonious causal graph without requiring explicit acyclicity constraints often imposed in static structural equation modeling.

Furthermore, this framework facilitates the interpretation of learned representations by mapping the sparse weight structures back to directed edges in a causal network [9]. The resulting models demonstrate superior performance in distinguishing direct causal links from indirect associations compared to standard attention mechanisms or unregularized deep learning baselines. By aligning the optimization objective with the statistical definition of Granger causality through gradient manipulation, these recurrent networks provide a scalable solution for uncovering nonlinear causal structures in sequential data, bridging the gap between flexible deep learning predictors and rigorous causal inference requirements.

3.1.3 Kernel-Based Nonlinear Mechanism Estimation

Kernel-based nonlinear mechanism estimation addresses the inherent limitations of linear Granger causality when applied to complex dynamical systems. By mapping time series data into high-dimensional reproducing kernel Hilbert spaces, these methods capture intricate nonlinear depen-



(a) Illustration of causal relations between attribute labels

Figure 10: Chart from 'Do learned representations respect causal relationships?' (arXiv:2204.00762)

dencies that traditional vector autoregressive models fail to detect. This approach effectively generalizes conditional independence testing, allowing for the identification of causal drivers in scenarios characterized by non-Gaussian noise and intricate functional forms without requiring explicit parametric assumptions about the underlying data generation process.

Recent advancements integrate kernel techniques with independent component analysis and neural architectures to enhance mechanism discovery. Algorithms such as Kernel Multilinear ICA utilize higher-order cumulants within a kernel framework to separate mixed causal sources, while other approaches employ kernel matrices to derive closed-form solutions for function class estimation. These hybrid strategies leverage the flexibility of kernel eigenvalue problems to model system dynamics, enabling the differentiation of direct causal influences from spurious correlations induced by latent confounders or immediate effects arising from slow sampling intervals.

Despite their theoretical robustness, kernel-based methods face computational challenges regarding scalability in high-dimensional settings due to the quadratic complexity of kernel matrix operations. Consequently, contemporary research focuses on optimizing these frameworks through sparse regression formulations and adaptive thresholding strategies to improve efficiency. By combining kernel mappings with gradient-based optimization or variational inference, modern estimators achieve a balance between modeling capacity and computational feasibility, providing a versatile platform for inferring causal structures in multivariate nonlinear systems where standard linear assumptions are violated.

3.2 Graph Neural Networks for Causal Structure Learning

3.2.1 Message Passing over Interventional Graph Topologies

Message passing over interventional graph topologies fundamentally alters the standard propagation mechanism by incorporating the structural mutilation induced by the do-operator. In this framework, intervening on a specific node necessitates the removal of all inbound edges to that node within the causal Directed Acyclic Graph (DAG), effectively severing its dependence on parent variables [10]. Consequently, message passing algorithms must dynamically adapt to these modified adjacency matrices, ensuring that information flow respects the truncated causal pathways rather than the original observational dependencies. This topological adjustment is critical for accurately estimating interventional distributions, as it prevents the leakage of spurious correlations from confounded paths that are blocked under intervention.

$$p(y \mid do(t)) = \sum_{m} p(m \mid t) \sum_{t'} p(t')p(y \mid m, t').$$

Figure 11: Chart from 'From Statistical to Causal Learning' (arXiv:2204.00607)

The implementation of such mechanisms often relies on neural architectures capable of processing both observational data and binary intervention indicators simultaneously. Models typically employ specialized layers, such as data-to-graph modules, to extract pairwise relational evidence and summarize it into graph messages that reflect the current interventional state. During the propagation phase, nodes aggregate messages only from their remaining parents in the mutilated graph, adhering to a strict topological order that preserves temporal causality. This ensures that the learned representations encode the specific causal effects of interventions, distinguishing between direct influences and indirect effects mediated by other nodes in the system.

Furthermore, advanced approaches integrate regularization techniques and attention mechanisms to refine the strength of causal edges within these interventional topologies. By minimizing loss functions designed to recover ground-truth causal structures from limited interventional samples, these methods enhance the robustness of the inferred graph against noise and unobserved confounding. The resulting message passing schemes not only facilitate the identification of Markov equivalence classes under intervention but also enable the precise calculation of counterfactual outcomes. Ultimately, this synthesis of graph neural networks with formal causal calculus allows for scalable and accurate inference in complex systems where traditional statistical methods fail to disentangle genuine causal relationships from mere associations.

3.2.2 Disentangling Causal and Spurious Edges via Dual Encoders

Disentangling causal mechanisms from spurious correlations is a fundamental challenge in representation learning, often addressed by architectures employing dual encoders. This paradigm typically utilizes two distinct encoding pathways: one dedicated to capturing invariant causal features that generalize across environments, and another designed to isolate environment-specific or spurious variations. By explicitly modeling the divergence between these latent subspaces, the framework enforces a structural separation where only the causal encoder informs the primary predictive task, while the spurious encoder absorbs confounding noise. This architectural bias ensures that learned representations adhere to underlying causal graphs rather than superficial statistical regularities present in the training distribution [9].

The optimization process relies on adversarial or contrastive objectives to maximize the independence between the causal and spurious latent variables. During training, the model minimizes reconstruction error while simultaneously penalizing mutual information between the two encoded representations, effectively forcing the disentanglement of factors. In specific implementations like CausalVAE, this is achieved by integrating a neural structural causal model that transforms independent exogenous variables into causally dependent latent factors through a linear causal layer [11]. Such mechanisms allow the network to simulate interventions, verifying that perturbations in the causal encoder propagate correctly to the output, whereas changes in the spurious encoder remain inert regarding the target variable.

Furthermore, this dual-encoder strategy enhances robustness against distributional shifts by ensuring that predictions rely solely on stable causal edges. Unlike standard deep learning models that may exploit shortcut learning via spurious correlations, these frameworks regularize the learning process to respect the directionality and modularity of the true data-generating process. The resulting adjacency matrix derived from the causal encoder reflects direct causal influences, filtering out indirect associations caused by common confounders. Consequently, this approach not only improves out-of-distribution generalization but also provides interpretable insights into the structural dependencies governing the observed data, bridging the gap between black-box neural networks and explicit causal discovery [2].

$$\sum_{s=1}^{2} \mathbf{T}_s(\mathbf{Ch}(\mathbf{x}))\boldsymbol{\lambda}_s(\mathbf{u}) = \sum_{s=1}^{2} \widetilde{\mathbf{T}}_s(\widetilde{\mathbf{C}}\widetilde{\mathbf{h}}(\mathbf{x}))\widetilde{\boldsymbol{\lambda}}_s(\mathbf{u}) + \mathbf{b}`,$$

Figure 12: Chart from 'CausalVAE: Structured Causal Disentanglement in Variational Autoencoder' (arXiv:2004.08697)

3.2.3 Adaptive Adjacency Learning for Spatiotemporal Dependencies

Adaptive adjacency learning addresses the limitations of static graph assumptions in spatiotemporal modeling by dynamically inferring latent dependencies from data. Unlike traditional methods relying on pre-defined physical topologies, these approaches treat the adjacency matrix as a learnable parameter optimized jointly with temporal dynamics. By leveraging gradient-based optimization and continuous relaxation techniques, models can discover directed acyclic graphs that reflect true causal mechanisms rather than mere correlations, effectively handling non-linear interactions and time-lagged effects inherent in complex systems.

Recent advancements integrate graph neural networks with structural learning modules to refine these adaptive structures iteratively. Architectures often employ dual-stream mechanisms where one branch distills relational evidence from high-dimensional observations into graph messages, while the other updates node embeddings based on the evolving topology. This end-to-end framework allows the model to prune spurious links and reinforce significant causal drivers, ensuring that the learned graph remains sparse and interpretable. Such methods are particularly effective in scenarios with missing data or latent confounders, as they can infer indirect paths and adjust edge weights to maximize likelihood under structural constraints.

The efficacy of adaptive learning lies in its ability to balance computational efficiency with representational fidelity across varying spatial scales. By incorporating regularization penalties that enforce acyclicity and stability, these models prevent overfitting to noise while capturing invariant causal relationships across different domains. Consequently, the resulting dynamic graphs provide a robust foundation for downstream tasks such as forecasting and intervention analysis, offering superior performance compared to fixed-structure baselines in environments characterized by shifting spatiotemporal dependencies.

3.3 Transformer and Attention Mechanisms in Causal Inference

3.3.1 Causal Attention Masks for Temporal Directionality

Causal attention masks serve as a critical architectural mechanism to enforce temporal directionality within deep learning models, ensuring that information flow adheres to the fundamental asymmetry of cause and effect. Unlike standard self-attention which permits bidirectional context aggregation, causal masking explicitly restricts the attention matrix to lower-triangular forms, preventing future time steps from influencing current representations. This structural constraint is essential for distinguishing genuine causal drivers from spurious correlations in time-series data, effectively embedding the principle that effects cannot precede their causes directly into the model's inductive bias.

Recent advancements have evolved beyond static triangular masks to incorporate dynamic, learnable masking strategies guided by estimated causal strengths. By integrating auxiliary loss functions derived from causal discovery metrics, these mechanisms refine attention weights to prioritize variables with high causal influence while suppressing non-causal dependencies. This approach allows the network to adaptively filter input features based on their temporal relevance, addressing limitations where vanilla attention fails to accurately communicate the relative importance of inputs or mistakenly attends to future contexts during training and inference.

Furthermore, the application of axial attention mechanisms combined with tied weight strategies enhances the scalability of causal masking in high-dimensional settings. These architectures decompose the attention operation along sample and variable axes, applying causal constraints specifically to the temporal dimension to maintain computational efficiency without sacrificing expressivity. Such designs facilitate robust causal representation learning in complex domains like motion forecasting and neural Granger causality, enabling models to capture non-linear delay equations and chaotic system dynamics while strictly preserving the directed nature of temporal interactions.

3.3.2 Cross-Attention Aggregation for Connectivity Estimation

Cross-attention aggregation mechanisms have emerged as a pivotal strategy for estimating connectivity within complex causal graphs, particularly by modeling interactions between distinct data points rather than solely within individual input-output pairs. Unlike traditional self-attention that operates on features of a single sample, cross-attention frameworks facilitate the exchange of information across the entire dataset, allowing the model to infer global structural dependencies. This approach enables the network to dynamically weigh the influence of specific temporal states or spatial nodes on one another, effectively capturing the nuanced directional flows required for accurate Granger causality estimation in high-dimensional time series.

In practical architectures, this aggregation is often implemented through dual-stream designs where data embeddings and graph structure estimates are processed concurrently. The cross-attention layers compute relation scores between hidden states of different variables, generating attention coefficients that serve as proxies for causal strength. These coefficients are subsequently aggregated, often via weighted summation, to construct intermediate connectivity matrices. To enhance robustness, recent methodologies integrate auxiliary causal graph discovery tasks, where the attention weights are regularized to align with theoretical causal constraints, thereby reducing the likelihood of learning spurious correlations driven by confounding factors.

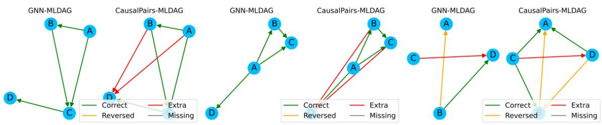
Furthermore, the efficiency of cross-attention aggregation is frequently optimized through tied weight mechanisms and axial attention strategies, which significantly reduce the computational complexity associated with pairwise comparisons in large-scale networks. By compressing data embeddings and applying reduction units, these models maintain scalability while preserving the fidelity of the estimated connectivity patterns. The resulting attention maps not only provide a differentiable approximation of the underlying directed acyclic graph but also offer interpretability, allowing researchers to visualize and validate the inferred causal pathways against domain-specific knowledge in fields ranging from neuroscience to financial modeling.

3.3.3 Interpretable Concept Bottlenecks via Structural Priors

Interpretable concept bottlenecks leverage structural priors to enforce semantic alignment between latent representations and human-understandable concepts, moving beyond opaque black-box embeddings. By integrating causal graphical constraints directly into the architecture, these models ensure that intermediate layers reflect genuine generative factors rather than spurious correlations. This approach typically employs continuous optimization techniques to learn Directed Acyclic Graphs (DAGs) over concept nodes, effectively restricting the hypothesis space to structurally valid causal mechanisms while maintaining differentiability for end-to-end training.

The incorporation of structural priors addresses the fundamental challenge of identifiability in high-dimensional settings where data scarcity often plagues traditional statistical methods. Unlike post-hoc explanation tools that approximate model behavior after training, structural bottlenecks embed interpretability as an inductive bias during the learning process. Methods utilizing amortized inference or neural structural causal models facilitate the disentanglement of variables by penalizing deviations from known physical laws or domain-specific dependency graphs. This ensures that the learned representation space adheres to predefined topological constraints, thereby enhancing robustness against distributional shifts and adversarial perturbations.

Furthermore, this paradigm shifts the focus from purely predictive accuracy to mechanistic transparency, allowing practitioners to intervene on specific concepts and observe downstream effects consistent with causal theory [12]. By decomposing complex predictions into a hierarchy of concept-based decisions governed by explicit structural equations, these models provide verifiable reasoning paths. Consequently, the resulting systems not only achieve competitive performance but also offer actionable insights into the underlying data generating process, bridging the gap between deep learning flexibility and the rigorous demands of scientific discovery.



(a) nonlin_simpson (b) symprod_simpson (c) mixed_simpson

Figure 13: Chart from 'From Observations to Causations: A GNN-based Probabilistic Prediction Framework for Causal Discovery' (arXiv:2507.20349)

4 Structural Causal Modeling and Graph Learning

4.1 Continuous Optimization for DAG Structure Learning

4.1.1 Trace Exponential Penalties for Acyclicity Enforcement

The trace exponential penalty serves as a foundational differentiable constraint for enforcing acyclicity in continuous causal structure learning. By leveraging the mathematical property that the trace of the matrix exponential of an adjacency matrix equals the sum of all cycle counts, this method transforms the discrete combinatorial problem of DAG validation into a smooth optimization landscape. Specifically, the function $h(\mathbf{A}) = \mathrm{tr}(e^{\mathbf{A} \circ \mathbf{A}}) - d$ evaluates to zero if and only if the graph contains no cycles, allowing gradient-based algorithms to simultaneously learn structural parameters and causal mechanisms without explicit discrete search procedures.

Despite its theoretical elegance, the direct computation of the matrix exponential presents significant numerical challenges, particularly as the dimensionality of the variable set increases. The operation is prone to floating-point overflow and exhibits cubic computational complexity, which can hinder scalability in high-dimensional settings. To mitigate these issues, practical implementations often employ matrix scaling techniques, utilizing the spectral radius of the initialized adjacency matrix as a safeguard, or restrict input values through clipping operations to maintain numerical stability during backpropagation.

Consequently, recent advancements have explored variants of the original trace exponential formulation to enhance efficiency and robustness. These include matrix power approximations that reduce computational overhead and algebraic characterizations based on the spectral radius, which offer linear or quadratic complexity bounds under specific low-rank assumptions. While the standard trace exponential remains a critical benchmark for ensuring global acyclicity in non-linear structural equation models, these optimized variants address the trade-offs between computational cost, numerical precision, and the strictness of the acyclicity constraint in large-scale applications.

4.1.2 Score-Based Search with Differentiable Relaxations

Score-based causal discovery traditionally relies on combinatorial optimization to maximize a scoring function, such as BIC or BDeu, over the discrete space of directed acyclic graphs (DAGs). However, the NP-hard nature of this search space limits scalability. To address this, recent methodologies employ differentiable relaxations that transform the discrete structural learning problem into a continuous optimization task. By representing the adjacency matrix as a set of continuous parameters, these approaches enable the use of efficient gradient-based solvers rather than heuristic discrete search strategies like greedy equivalence search or integer programming.

The core innovation in this domain involves enforcing the acyclicity constraint through smooth, differentiable penalties rather than explicit combinatorial checks. A seminal approach utilizes a trace exponential formulation to characterize acyclicity, allowing the constraint to be incorporated directly into the loss function as a regularization term. This formulation permits the simultaneous optimization of graph structure and model parameters via standard backpropagation. Consequently, the search procedure avoids the computational bottlenecks associated with evaluating scores for invalid cyclic graphs, significantly accelerating convergence while maintaining theoretical guarantees regarding the validity of the resulting DAG.

Furthermore, advanced variants integrate neural network architectures to model complex, non-linear functional relationships between variables within this relaxed framework. These methods often utilize Gumbel-Softmax reparameterization or sigmoid activations to approximate discrete edge existence probabilities, facilitating end-to-end training. While effective, these techniques require careful tuning of hyperparameters balancing the data fit score against the acyclicity penalty to prevent convergence to local minima or dense, spurious graphs. Despite these challenges, differentiable relaxations have established a new paradigm for scalable structure learning, bridging the gap between classical score-based statistics and modern deep learning optimization.

4.1.3 Topological Sorting via Spectral Decomposition

Topological sorting via spectral decomposition offers a continuous relaxation for discovering causal orderings in directed acyclic graphs (DAGs) by leveraging the spectral properties of graph Laplacians or adjacency matrices [13]. Unlike discrete combinatorial search methods that suffer from super-exponential complexity, this approach embeds nodes into a low-dimensional spectral space where the relative positions of eigenvectors encode potential causal directions. By analyzing the eigenvalues and corresponding eigenvectors of the graph's structural matrix, one can derive a soft ordering that approximates the strict topological constraints required for acyclic structures, effectively transforming a discrete permutation problem into a differentiable optimization task.

The core mechanism relies on the observation that for a valid topological ordering, the associated adjacency matrix can be permuted into a strictly upper triangular form. Spectral decomposition facilitates this by identifying a permutation matrix that minimizes the mass of the lower triangular part of the reordered matrix, often formulated as minimizing a trace objective involving the Laplacian. Techniques frequently employ the Fiedler vector or higher-order eigenvectors to sort nodes, ensuring that parents precede children in the inferred sequence. This spectral embedding is particularly robust against noise, as the global structural information captured by dominant eigenmodes tends to preserve the underlying causal hierarchy even when local edge estimates are uncertain.

Furthermore, integrating spectral sorting with continuous constraint optimization allows for end-to-end learning of graph structures without explicit enumeration of orderings. Recent advancements combine spectral initialization with gradient-based methods to refine the adjacency matrix while enforcing acyclicity through smooth penalties, such as trace exponential formulations. This hybrid strategy mitigates the risk of converging to local minima common in purely score-based approaches. By decoupling the ordering inference from edge weight estimation, spectral decomposition provides a computationally efficient prior that significantly narrows the search space, enabling scalable causal discovery in high-dimensional settings where traditional constraint-based algorithms become intractable.

4.2 Integration of Large Language Models in Causal Discovery

4.2.1 Semantic Variable Abstraction and Factor Proposal

Semantic variable abstraction addresses the critical trade-off between precision and granularity by constructing high-level representations from raw observational data. Rather than relying solely on primitive node features, this approach synthesizes meaningful variables through graph schemas, meta-paths, and domain-specific semantics. By encapsulating fundamental causal mechanisms such as collision or support within primitive types, the methodology ensures semantic consistency across compositions, allowing models to dynamically construct explanations across multiple levels of abstraction.

Factor proposal extends this abstraction by identifying latent causal factors that govern the composition of observed variables, often termed diminutive causal structures. Instead of treating features as atomic units, the process decomposes them into linear combinations of underlying factor variables, effectively mapping the search space from individual edges to higher-order interactions. This facilitates the discovery of c-factors and ancestor sets, enabling the application of factorization theorems to sub-graphs under proper interventions and providing a robust alternative to traditional symbolic reasoning.

The integration of these abstractions into neural architectures allows for the amortized inference of causal structures without explicit constraint enumeration. By designating causal matrices as function parameters, the model learns to prune redundant connections and isolate Markov blanket representations, effectively blocking task-irrelevant information. This end-to-end training regime refines latent features up to equivalence relations, ensuring that the resulting causal graphs capture persistent dependencies and orientation asymmetries while maintaining computational efficiency and interpretability [14].

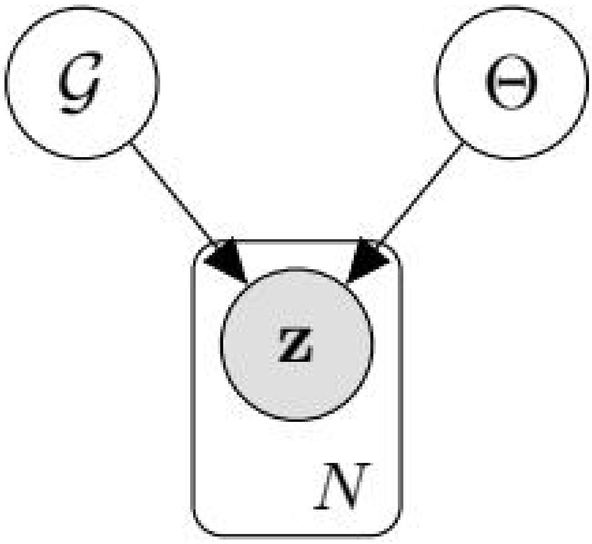


Figure 14: Chart from 'LEARNING LATENT STRUCTURAL CAUSAL MODELS' (arXiv:2210.13583)

4.2.2 Knowledge-Guided Skeleton Refinement and Pruning

Knowledge-guided skeleton refinement leverages external domain expertise to constrain the search space of causal structure learning, effectively transforming an ill-posed problem into a tractable optimization task. By integrating prior knowledge graphs or expert-defined constraints, algorithms can enforce specific adjacency requirements or forbid implausible connections before the primary learning phase. This approach significantly reduces the combinatorial complexity associated with discovering the undirected skeleton, ensuring that the resulting graph topology aligns with established scientific principles while maintaining flexibility for data-driven discovery of novel relationships.

Following the initial skeleton identification, pruning mechanisms are employed to eliminate spurious edges that arise from statistical noise or finite sample biases. Advanced techniques utilize adaptive thresholding on weighted adjacency matrices, where edge weights below a dynamically determined cutoff are severed to enforce sparsity. Recent differentiable frameworks incorporate these pruning operations directly into the loss function via regularization terms, such as L1-norm penalties or acyclicity constraints, allowing the model to iteratively refine the skeleton through gradient-based optimization rather than relying on discrete, post-hoc processing steps that may disrupt global optimality.

The synergy between knowledge guidance and algorithmic pruning yields substantial improvements in structural accuracy, particularly in high-dimensional regimes with limited data. Empirical evidence suggests that combining topological constraints with iterative refinement strategies markedly reduces the Structural Hamming Distance compared to unconstrained baselines. Furthermore, this hybrid methodology enhances the robustness of causal discovery against distributional shifts, as the preserved skeleton retains only those dependencies that are both statistically significant and semantically consistent with the injected prior knowledge, thereby facilitating more reliable downstream causal inference.

4.2.3 Hybrid Statistical and Symbolic Reasoning Frameworks

Hybrid statistical and symbolic reasoning frameworks address the limitations of purely data-driven or logic-based approaches by integrating the scalability of neural networks with the interpretability of symbolic systems. Traditional constraint-based and score-based methods often struggle with high-dimensional data or fail to distinguish structures within Markov equivalence classes without strong assumptions like faithfulness. By embedding symbolic constraints, such as acyclicity or specific functional forms, directly into differentiable learning objectives, these hybrid models leverage gradient-based optimization to discover causal graphs while adhering to rigorous logical formalizations.

A core advantage of this paradigm is the ability to recover explicit causal mechanisms rather than mere associative patterns [15]. Recent architectures incorporate symbolic trees or closed-form equations attached to graph edges, enabling the extraction of analytical Jacobians and facilitating compatibility with established causal inference tools. This synthesis allows for modular training on complex datasets, including those requiring front-door adjustments, where deep neural networks capture intricate non-linear dependencies while symbolic components ensure the resulting structure remains logically consistent and human-interpretable.

$$\mathcal{F}_2(\boldsymbol{\theta}) = \lambda_2 \sum_{i<j} (\|\boldsymbol{\theta}_{ij}\|_F - \tau)_+,$$

Figure 15: Chart from 'A Unified Framework for Structure-Aware Clustering and Heterogeneous Causal Graph Learning' (arXiv:2605.19313)

Despite promising theoretical foundations, developing scalable causal reasoning engines remains an open challenge, particularly regarding generalizability across domains. Current research focuses on amortizing the learning of topological orders and utilizing reinforcement learning to navigate the combinatorial space of structural hypotheses more efficiently. By bridging continuous simulation with discrete causal explanation, these frameworks aim to overcome the brittleness of standard deep learning models, offering robust solutions that maintain causal validity even under distributional shifts or limited interventional data.

4.3 Causal Abstraction and Model Reduction

4.3.1 Category-Theoretic Mappings between Micro and Macro Models

This section formalizes the relationship between fine-grained micro-level structural causal models and coarse-grained macro-level abstractions using category-theoretic functors. By treating causal structures as objects within a category and causal mappings as morphisms, we establish a rigorous framework for analyzing how local mechanistic interactions aggregate into global system behaviors. This approach moves beyond ad-hoc reductionism, providing a mathematically sound basis for defining when a macro-model faithfully represents the underlying micro-dynamics without losing essential causal information.

The core of this mapping relies on constructing functors that preserve the compositional structure of causal mechanisms across scales. Specifically, we examine how deterministic and stochastic transitions in micro-SCMs can be mapped to effective operators in macro-models while maintaining identifiability conditions. These functors ensure that interventional distributions at the macro level correspond consistently to sets of interventions at the micro level, thereby validating the use of simplified models for policy learning and counterfactual reasoning in complex, high-dimensional environments.

Furthermore, this categorical perspective elucidates the conditions under which emergent properties arise that are not explicitly present in the constituent micro-components. By analyzing the limits of these mappings, we identify scenarios where standard aggregation methods fail to capture non-linear dependencies or feedback loops inherent in the primitive hierarchy. This theoretical grounding facilitates the development of scalable inference algorithms that leverage micro-data to constrain macro-parameter spaces, ultimately enhancing both the interpretability and generalization capabilities of hierarchical causal models.

4.3.2 Constructive Transformations for Interventional Consistency

Constructive transformations for interventional consistency rely on the principle that valid causal mechanisms must remain invariant under specific structural manipulations, ensuring that predicted distributions align with actual interventional outcomes. Unlike observational equivalence classes where Markov equivalent structures yield identical statistical dependencies, interventional consistency demands that the model's internal representation explicitly encodes the directionality and modularity of causal relationships. This requires transforming the learned representation such that the application of a do-operator corresponds to a well-defined modification of the structural equations, preserving the autonomy of unaffected mechanisms while accurately propagating changes through the graph.

A primary technique involves leveraging interchange interventions to enforce alignment between high-level causal variables and low-level neural representations. By swapping specific latent features between different input samples and comparing the model's output against the expected counterfactual outcome, these transformations penalize representations that fail to isolate independent causal factors. This process effectively disentangles the underlying generative process, ensuring that the model does not rely on spurious correlations that vanish under intervention. Consequently, the learning objective shifts from minimizing predictive error on static data to satisfying consistency constraints across a spectrum of hypothetical manipulations, thereby grounding the model in the true data-generating process.

Furthermore, these transformations utilize statistical divergences between observational and interventional distributions to infer causal ordering and validate structural assumptions. Under the assumption of interventional faithfulness, any detectable distributional shift resulting from an intervention must correspond to a directed path in the underlying graph. Constructive methods exploit this property by iteratively refining the causal graph structure to minimize the discrepancy between the model's implied interventional distributions and the empirical data obtained from hard or soft interventions. This approach not only enhances identifiability in the presence of latent confounders but also ensures that the resulting causal model generalizes robustly to unseen domains and novel intervention sets [16].

$$C_j = \frac{\mathbb{P}(x_j)\mathbb{P}(L|x_j)}{\sum_i^N \mathbb{P}(x_i)\mathbb{P}(L|x_i)} x_j$$

Figure 16: Chart from 'A Review and Roadmap of Deep Causal Model from Different Causal Structures and Representations' (arXiv:2311.00923)

4.3.3 Hierarchical Primitive Composition for Dynamic Systems

Hierarchical primitive composition addresses the limitations of monolithic causal models in dynamic systems by decomposing complex mechanisms into reusable, interpretable units. This approach constructs a multi-layered architecture where low-level physical dynamics primitives are aggregated into higher-level functional abstractions. By enforcing structural minimality, the framework replaces subgraphs of elementary interactions with single composite primitives whenever their collective input-output behavior matches a predefined signature, thereby reducing computational complexity while preserving causal fidelity.

The integration of these primitives relies on a modular training strategy grounded in topological ordering and constraint satisfaction. Algorithms sequentially recover latent components starting from root nodes, utilizing pruning subroutines to eliminate extraneous edges and refine the underlying Directed Acyclic Graph. This process ensures that the composition of primitives adheres to strict temporal tiers and conditional independence constraints, effectively navigating the search space to identify identifiable causal structures even under incomplete temporal information or missing data scenarios.

Furthermore, this hierarchy facilitates the bridge between continuous neural representations and discrete symbolic reasoning. By encoding system states into latent codes that decode into causal graphs, the model supports dynamic adaptation where belief states are shaped by generative environment models. The resulting architecture enables the recovery of nonlinear mechanisms through closed-form equations and analytical Jacobians, offering a robust solution for modeling time-varying systems that require both precise mechanism recovery and the flexibility to handle evolving causal relationships without succumbing to the ill-posed nature of standard neural weight spaces.

5 Future Directions

Despite the significant theoretical advancements synthesized in this survey, current methodologies in causal representation learning face persistent limitations that hinder their widespread deployment in real-world applications. A primary bottleneck remains the reliance on strong, often unrealistic assumptions regarding the nature and availability of interventions. Many frameworks require explicit knowledge of intervention targets or assume access to multiple distinct environments with sufficient distributional overlap, conditions that are rarely met in complex, high-dimensional domains such as healthcare or autonomous systems. Furthermore, while identifiability guarantees have been established under specific parametric constraints, these often degrade when applied to noisy, finite datasets where the assumed sparsity or linearity of causal mechanisms does not strictly hold. The computational complexity of optimizing these models also scales poorly with dimensionality, frequently leading to latent space collapse or convergence to suboptimal local minima that fail to capture the true underlying causal graph.

Future research must prioritize the development of more robust and flexible frameworks that relax these stringent requirements while maintaining theoretical rigor. One promising direction involves leveraging self-supervised learning paradigms to automatically discover latent intervention targets from purely observational data, thereby reducing the dependency on explicit environmental labels or controlled experimental settings. Investigating hybrid approaches that combine temporal nonstationarity with weak structural priors could further enhance identifiability in scenarios where hard interventions are impossible. Additionally, there is a critical need to design scalable algorithms capable of handling continuous, high-dimensional data streams without sacrificing the interpretability of the learned representations. This includes exploring neural architectures that inherently encode causal inductive biases, such as modular networks with dynamic connectivity, to better approximate the sparse mechanism shifts observed in natural systems. Bridging the gap between abstract causal theory and practical deep learning optimization will require novel regularization techniques that stabilize training dynamics while enforcing causal consistency.

The successful realization of these future directions holds the potential to fundamentally transform the landscape of artificial intelligence by enabling systems that possess genuine reasoning capabilities rather than mere pattern matching. By achieving robust causal disentanglement, next-generation models will be able to generalize reliably under severe distributional shifts, ensuring safety and reliability in critical applications where failure is not an option. Moreover, the ability to perform accurate counterfactual reasoning will empower AI systems to support fair decision-making by explicitly identifying and mitigating biases rooted in confounding variables. Ultimately, integrating causal principles into deep generative models will pave the way for interpretable agents capable of explaining their decisions, adapting to novel environments with minimal data, and engaging in complex planning tasks that require an understanding of cause and effect, marking a decisive step toward artificial general intelligence.

6 Conclusion

This survey has systematically synthesized the convergence of causal inference and deep representation learning, highlighting how modern methodologies overcome the inherent unidentifiability of standard black-box models. We established that leveraging environmental variability, whether through multi-domain shifts or temporal nonstationarity, provides the critical inductive biases necessary to recover latent causal structures without explicit supervision. The review distinguished between the rigorous constraints imposed by hard interventions and the flexible, albeit theoretically challenging, nature of soft interventions, demonstrating how each shapes the identifiability of underlying graphs. Furthermore, we detailed how enforcing sparsity regularization and adopting the principle of Independent Causal Mechanisms effectively break rotational symmetries, enabling the disentanglement of content from style through causal abstraction. Finally, the integration of these frameworks into variational inference pipelines was shown to facilitate robust counterfactual reasoning and fairness, bridging the gap between theoretical guarantees and practical generative capabilities in high-dimensional spaces.

The significance of this comprehensive review lies in its provision of a unified taxonomy that categorizes diverse approaches based on their reliance on intervention types, structural assumptions, and dynamic constraints. By critically evaluating the trade-offs between model flexibility and identifiability guarantees, this work clarifies the conditions under which deep generative models can transition from capturing spurious correlations to learning invariant causal mechanisms. This synthesis is pivotal for advancing artificial intelligence beyond mere pattern recognition, offering a foundational roadmap for developing systems capable of robust generalization under distributional shifts. The insights presented here underscore the necessity of embedding causal semantics into representation learning to ensure that AI systems are not only accurate but also interpretable, manipulable, and reliable in safety-critical domains such as healthcare and autonomous decision-making.

Looking forward, the field must address the remaining challenges in scaling these causal frameworks to complex, real-world datasets where intervention targets are unknown and causal topologies are intricate. Future research should prioritize the development of algorithms that can automatically detect and exploit subtle distributional shifts without relying on strong parametric assumptions or perfect intervention knowledge. There is an urgent need to refine optimization strategies that prevent latent space collapse while maintaining the sparsity and modularity required for true disentanglement. We call upon the research community to foster deeper interdisciplinary collaboration between causal theorists and deep learning practitioners to create next-generation architectures that inherently support counterfactual reasoning. By rigorously pursuing these directions, we can realize the vision of artificial intelligence that possesses a genuine understanding of the generative processes governing our world, ultimately leading to more trustworthy and equitable technological advancements.

References

[1] Towards Interpretable Deep Generative Models via Causal Representation Learning

[2] Group Interventions on Deep Networks for Causal Discovery in Subsystems

[3] Learning Structural Causal Models through Deep Generative Models Methods, Guarantees, and Challenges

[4] Navigating causal deep learning

[5] INTERPRETABLE CAUSAL REPRESENTATION LEARNING FOR BIOLOGICAL DATA IN THE PATHWAY SPACE

[6] Disentangled Representation Learning

[7] Causal Manifold Fairness Enforcing Geometric Invariance in Representation Learning

[8] Causal Analysis of Customer Churn Using Deep Learning

[9] Do learned representations respect causal relationships

[10] FROM STATISTICAL TO CAUSAL LEARNING

[11] CausalVAE Structured Causal Disentanglement in Variational Autoencoder

[12] From Observations to Causations A GNN-based Probabilistic Prediction Framework for Causal Discovery

[13] Causal Order Discovery based on Monotonic SCMs

[14] LEARNING LATENT STRUCTURAL CAUSAL MODELS

[15] DAG-based Penalized Clustering

[16] A Review and Roadmap of Deep Causal Model from Different Causal Structures and Representations